Lab 6 ticket

Fit and Compress — Friday, December 4

Lab 6 — Fit and Compress · Fri Dec 4 · ~20 min (SVD lecture + project launch day) · Laptop · Handwrite · Initials

Name: _____ **Language (circle):** Octave / Python

New syntax

I needed to…	Command / pattern
Least squares	
SVD	
Rank-k rebuild	
Frobenius residual	

C6.0. $\hat{c} =$ _____ $\|A\hat{c} - b\| =$ _____ **Expected:** $\approx (1.167, 0.5)$, residual ≈ 0.408.

C6.0b. SVD of A: left singular vectors look orthonormal? Y / N

C6.1. Singular values of M: _____

C6.2. $\|M - M_1\|_F =$ _____ $\|M - M_2\|_F =$ _____

C6.3. Storage $k(m + n + 1)$ for this 4×4: $k = 1 \rightarrow$ _____ $k = 2 \rightarrow$ _____
For 100×100, k with $k(201) < 5000$: $k \leq$ _____

Stuck on: _____

Check-in initials: _____